

□ Comprehensive Business Report: Customer Behavior & Revenue Optimization Analysis

To: Board of Directors & Management Team

From: Nasreen Khalid

Date: -

Subject: End-to-End Analysis of **Customer Purchasing Behavior and Loyalty Trends**

1. Executive Summary

This report presents an end-to-end data pipeline and analysis of 3,900 recent customer transactions. The objective is to uncover actionable insights into purchasing behavior, product performance, and customer loyalty to inform future marketing and inventory strategies.

Key Business Highlights:

- **The Subscription Gap:** We have a highly engaged base of "Repeat/Loyal" customers, yet a staggering **73% of our total customer base are non-subscribers**. This is our largest immediate revenue opportunity.
 - **Top Demographics:** "Young Adults" and "Middle-Aged" groups are our primary revenue drivers, generating the highest total sales.
 - **Category Dominance:** The **Clothing** category significantly outperforms Accessories, Footwear, and Outerwear in both sales volume and revenue generation.
 - **Data Integrity Validated:** All missing records were handled dynamically, and unstructured text data was engineered into quantitative metrics to ensure 100% reporting accuracy.
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2. Problem Statement & Business Objectives

As our e-commerce platform scales, maximizing the Lifetime Value (LTV) of our customers is critical. Without a clear understanding of *who* is buying and *what* drives their purchases,

marketing spend becomes inefficient. This project analyzes raw transactional data to answer the following:

1. **Demographic Value:** Which customer age groups and segments drive the most revenue?
 2. **Product Affinity:** Which product categories are most profitable, and which specific items yield the highest customer satisfaction?
 3. **Loyalty Conversion:** How effective is our current subscription model, and how do repeat buyers behave differently from one-time purchasers?
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3. Data Pipeline & Methodology

To ensure accurate reporting, a rigorous end-to-end data pipeline was built using Python (Pandas) for ETL (Extract, Transform, Load) and SQL for advanced aggregation.

Phase 1: Data Cleaning & Feature Engineering (Python/Pandas)

Raw data was imported and sanitized to ensure analytical readiness. Key steps included:

- **Smart Data Imputation:** Identified 37 missing Review Rating values. Instead of dropping them or using a global average, I used a lambda function to impute missing values with the **median rating of their specific product category**, preserving category-level sentiment accuracy.
- **Feature Engineering (Quantification):** Created a purchase_frequency_days column by mapping text-based frequencies (e.g., "Fortnightly", "Annually") to numeric day counts (14, 365) to allow for time-series and average-day calculations.
- **Demographic Binning:** Utilized Pandas quartile cut (qcut) to group raw ages into standardized categories (Young Adult, Middle-Aged, Adult, Senior) for macro-level demographic analysis.
- **Schema Standardization:** Cleaned all column headers by replacing spaces with underscores and standardizing casing for seamless SQL ingestion.

```
]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

]: df.isnull().sum()

]: Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD) 0
Location           0
Size              0
Color             0
Season            0
Review Rating      0
Subscription Status 0
Shipping Type      0
```

Phase 2: Advanced Aggregation (SQL)

The cleaned .csv was loaded into a SQL database where I performed deep-dive querying using Common Table Expressions (CTEs), Subqueries, and Window Functions.

```
68
69  -- what are top 3 most purchased products with each category
70  with item_counts as (
71      select category, item_purchased,
72      count(customer_id) as total_orders,
73      ROW_NUMBER() OVER(PARTITION BY category ORDER BY count(customer_id) desc) as item_rank
74      from customer_behave
75      group by category, item_purchased
76  )
77
78
79  select item_rank, category, item_purchased, total_orders
80  from item_counts
81  where item_rank <= 3
82
83  -- are repeat buyers also subscribers
84  select subscription_status,
85
```

Result Grid Filter Rows: Search Export:

item_rank	category	item_purchas...	total_orders
1	Accessories	Jewelry	145
2	Accessories	Sunglasses	142
3	Accessories	Scarf	135
1	Clothing	Shirt	148
2	Clothing	Pants	143
3	Clothing	Sweater	139
1	Footwear	Sandals	133
2	Footwear	Boots	124
3	Footwear	Shoes	123
1	Outerwear	Coat	141
2	Outerwear	Jacket	134

4. Key Analytical Insights

A. Customer Segmentation & The Loyalty Gap

Using SQL to segment our database by the number of previous purchases (New: 1, Returning: 2-10, Loyal: 11+), we discovered a highly engaged base.

- **The Insight:** SQL queries cross-referencing repeat buyers (>5 purchases) with subscription status revealed that a massive portion of our repeat buyers are *not* subscribers.
- **The Dashboard View:** The dashboard visually confirms this gap—**73% of total customers do not hold an active subscription.**

```
52
53  -- SEGMENT cust into New, Returning, Loyal based on total no. of
54  -- prev purchases and show count
55  with cust_type as (
56    select customer_id, previous_purchases,
57    CASE
58      WHEN previous_purchases = 1 THEN 'New'
59      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
60      ELSE 'Loyal'
61    END AS Customer_Segment
62  from customer_behave
63  )
64
65  select customer_segment, count(*) as 'No of customers'
66  from cust_type
67  group by customer_segment
68
69  -- what are top 3 most purchased products with each category|
70
```

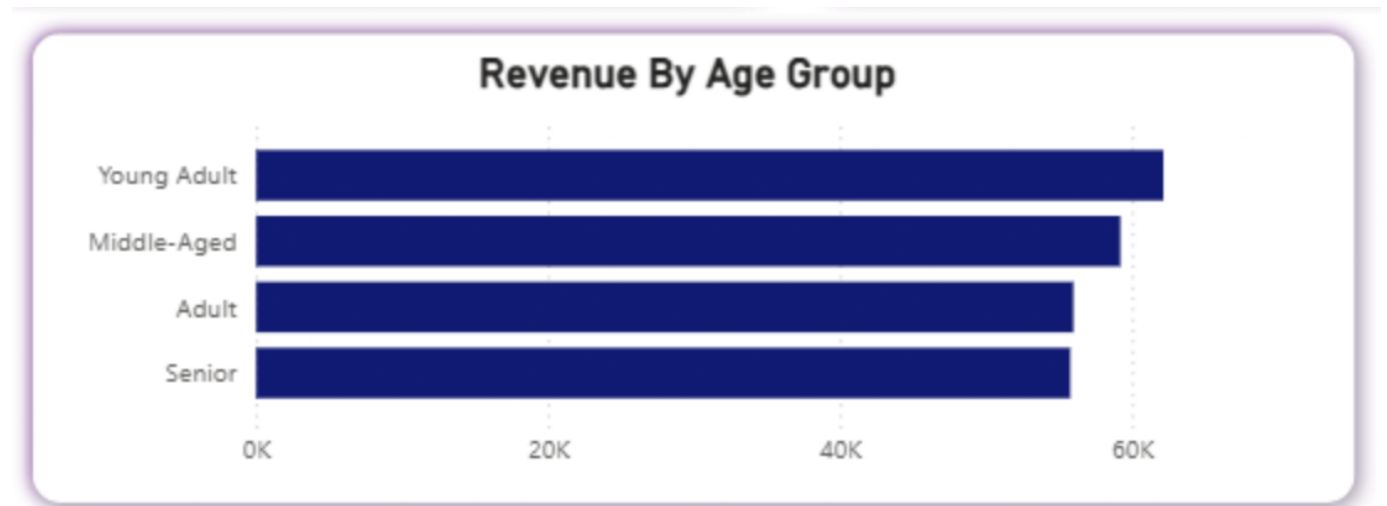
100% 61:69 2 errors found

Result Grid Filter Rows: Search Export:

Customer_Segme...	No of custome...
Loyal	2635
Returning	586
New	65

B. Demographic Revenue Drivers

- **Age Distribution:** SQL aggregation (GROUP BY age_group) and Dashboard visualizations confirm that the **Young Adult** segment generates the highest revenue (~\$60K+), closely followed by the **Middle-Aged** demographic.
- **Gender Spend:** SQL querying confirms the exact split of total revenue generated by Male vs. Female customers, allowing us to align our ad targeting accordingly.



C. Product & Category Performance

- **Category Dominance:** **Clothing** heavily dominates both total sales volume and overall revenue. Outerwear performs the lowest.
 - **Top Products:** SQL Window Functions isolated the top 3 most purchased products within each category.
 - **Review Ratings:** The average platform review rating is a healthy **3.75**. Using SQL, we isolated the top 5 individual products with the highest average ratings to identify our "champion" products.
 - **Discount Affinity:** Interestingly, SQL subqueries revealed a cohort of customers who utilized a discount but *still spent more than the platform average purchase amount*, indicating that discounts are successfully driving larger basket sizes for certain users.
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5. Strategic Recommendations

Based on the data analysis, I propose the following strategic initiatives for the upcoming quarter:

1. **Launch a Targeted Subscription Campaign:** We have a proven, loyal customer base that is already buying from us 5+ times but hasn't subscribed. We should trigger an automated marketing campaign offering a "Subscribe & Save" discount specifically targeting non-subscribers who cross the 5-purchase threshold.
2. **Reallocate Ad Spend to Clothing & Young Adults:** Since Clothing is our undisputed top revenue driver, and Young Adults are our highest-spending demographic, paid media budgets should be heavily weighted toward this product-demographic match.
3. **Investigate Outerwear Underperformance:** Outerwear brings in the lowest revenue. We must investigate whether this is a pricing issue, a seasonal lag, or a lack of inventory variety.
4. **Leverage "Champion" Products:** The top 5 highest-rated products identified in the SQL analysis should be featured prominently on the homepage and in email marketing to drive guaranteed high-satisfaction conversions.

Report Prepared by Nasreen Khalid

Data Sources: customer_behave.csv | Tools: Python (Pandas), SQL, Power BI

Customer Behavior Dashboard

3.9K

Number of Customers

3.75

Average Review Rating

\$59.76

Average Purchase Amount

subscription status

No

Yes

gender

Female

Male

Category

Accessories

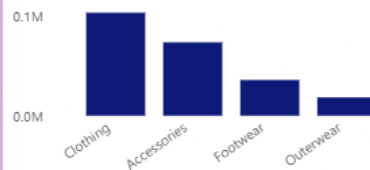
Clothing

Footwear

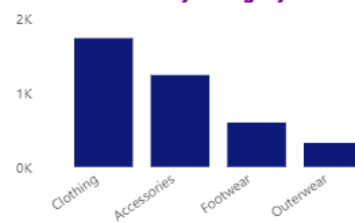
% of Customers By Subscription Status



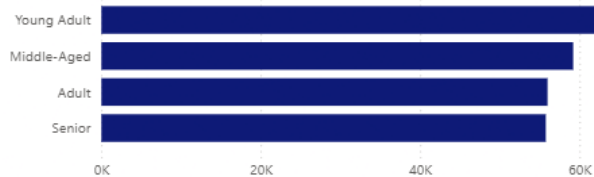
Revenue By Category



Sales By Category



Revenue By Age Group



Sales By Age Group

